Principles of Statistics

Part II

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Principles of Statistics

0. INTRODUCTION

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $X : \Omega \to \mathbb{R}$ a random variable with distribution function

$$F(t) = \mathbb{P}\big(\{\omega : X(\omega) \leq t\}\big) = \begin{cases} \displaystyle\int_{-\infty}^{t} f(x)\,dx, & \text{if } X \text{ is continuous with pdf } f, \\ \displaystyle\sum_{x \leq t} f(x), & \text{if } X \text{ is discrete with pmf } f. \end{cases}$$

We denote $P = \mathbb{P} \circ X^{-1}$, the law or "distribution" of X.
We observe $X_1, \ldots, X_n$, which are iid copies of X, where n is the sample size.

Def 1. A statistical model is a collection $\{f_\theta : \theta \in \Theta\}$ of pdfs or pmfs, or $\{P_\theta : \theta \in \Theta\}$ of laws, indexed by a parameter space Θ.

Ex.

- $N(\theta, 1)$, $\Theta = \mathbb{R}$.
- $N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$, $\Theta = \mathbb{R} \times (0, \infty)$.
- $\text{Poi}(\theta)$, $\Theta = [0, \infty)$.

Def 2. We say the model is correctly specified when $\exists\, \theta_0 \in \Theta$ such that $P = P_{\theta_0}$.
We will refer to the "true parameter" as θ_0.

Goals of Statistics

1. Estimation: We define an estimator
$$\widehat{\theta}_n = \widehat{\theta}_n(X_1, \ldots, X_n),$$
which we hope is close to θ_0 when $X_i \overset{\text{iid}}{\sim} P_{\theta_0}$, for all $\theta_0 \in \Theta$, when n is large.
2. Testing: We have a null hypothesis
$$H_0 : \theta \in \Theta_0 \subseteq \Theta,$$
and an alternative hypothesis
$$H_1 : \theta \in \Theta_1 \subseteq \Theta.$$
We define a test,
$$\psi_n := \psi_n(X_1, \ldots, X_n),$$
such that $\psi_n = 0$ if H_0 is true, and $\psi_n = 1$ if H_1 is true, w.h.p.

3. Uncertainty Quantification

- Define a confidence set $C_n(X_1, \ldots, X_n; \alpha)$, such that

$$P_\theta(\theta \in C_n) \overset{(\geq)}{=} 1 - \alpha \qquad \forall\, \theta \in \Theta,$$

  where α is the significance level.
- Define a Bayesian credible set, a region of high posterior probability.

I. LIKELIHOOD PRINCIPLE

Def 3. The likelihood function

$$L_n(\theta) = \prod_{i=1}^{n} f_\theta(X_i).$$

Note: This is a random function, as it depends implicitly on $X_1, \ldots, X_n$.

The log-likelihood function

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f_\theta(X_i),$$

we write

$$\overline{\ell}_n(\theta) = \frac{1}{n}\ell_n(\theta)$$

for a normalised log-likelihood.

Def 4. A maximum likelihood estimator $\widehat{\theta}_{\text{MLE}}$ is any estimator with

$$L_n\big(\widehat{\theta}_{\text{MLE}}\big) = \max_{\theta \in \Theta} L_n(\theta).$$

Ex.

$$X_i \overset{\text{iid}}{\sim} \text{Poi}(\theta).$$

$$L_n(\theta) = \prod_{i=1}^{n} e^{-\theta} \frac{\theta^{X_i}}{X_i!}.$$

$$\ell_n(\theta) = -n\theta + (\log \theta) \sum X_i + \text{const.}$$

$$\ell_n'(\theta) = -n + \frac{\sum X_i}{\theta} \overset{\text{FOC}}{=} 0$$

$$\implies \quad \text{suggests } \widehat{\theta}_{\text{MLE}} = \frac{\sum_{i=1}^{n} X_i}{n} := \overline{X}_n \qquad \text{“sample mean”}.$$

SOC:

$$\ell_n''(\theta) = -\frac{1}{\theta^2} \sum X_i < 0 \qquad \text{if } \sum X_i > 0.$$

If $\sum X_i = 0$,

$$\ell_n(\theta) = -n\theta \quad \implies \quad \widehat{\theta}_{\text{MLE}} = 0.$$

In every case

$$\widehat{\theta}_{\text{MLE}} = \overline{X}_n.$$

Ex.

$$X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \theta = (\mu, \sigma^2).$$

$$\ell_n(\theta) = -\frac{n}{2}\log\pi - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

FOC:

$$\frac{\partial \ell_n}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0,$$

$$\frac{\partial \ell_n}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0.$$

We can check that the Hessian of ℓ_n is negative definite $\forall\, \theta \in \mathbb{R} \times (0, \infty)$.
Hence,

$$\widehat{\theta}_{\text{MLE}} = \begin{pmatrix} \overline{X}_n \\ \underbrace{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2}_{\text{“Sample variance”}} \end{pmatrix}.$$

Non-examinable

The likelihood principle poses that if two observations

$$X = (X_1, \ldots, X_n) \qquad \text{and} \qquad X^* = (X_1^*, \ldots, X_n^*)$$

give rise to likelihoods L_n, L_n^* respectively, such that

$$L_n(\theta) = L_n^*(\theta) \cdot M \qquad \forall\, \theta \in \Theta,$$

and some constant M, then they should produce the same “inference” on the true parameter.

- The MLE, Bayes estimators, credible sets satisfy the likelihood principle.
- Confidence sets, tests generally do not.

Birnbaum (1962)

$$\text{Conditionality} + \text{Sufficiency} \quad \Longrightarrow \quad \text{Likelihood principle.}$$

End of lecture 1

II. Information Geometry

The score function is

$$S_n(\theta) = \frac{\partial}{\partial \theta}\ell_n(\theta), \qquad \Theta \subseteq \mathbb{R}^d.$$

In many cases,

$$S_n(\widehat{\theta}_{\text{MLE}}) = 0,$$

where, by convention,

$$S_n(\theta') = \frac{\partial}{\partial \theta}\ell_n(\theta)\bigg|_{\theta=\theta'}.$$

Let X be a r.v. taking values in $\mathcal{X}$, and $g : \mathcal{X} \to \mathbb{R}$. Take a model $\{f(x,\theta) : \theta \in \Theta\}$. We will write

$$\mathbb{E}_\theta g(X) = \int g(x) f(x,\theta)\, dx, \qquad X \sim f(\cdot,\theta).$$

Obs 1.

Suppose the model $\{f(\cdot,\theta) : \theta \in \Theta\}$ is such that

$$\mathbb{E}_{\theta_0} \big|\log f(X,\theta)\big| < \infty \qquad \forall \theta \in \Theta.$$

Define

$$\ell(\theta) = \mathbb{E}_{\theta_0} \log f(X,\theta), \qquad X \sim P_{\theta_0}.$$

The normalised log-likelihood

$$\bar{\ell}_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log f(X_i,\theta)$$

can be thought of as a sample version of $\ell(\theta)$.

Suppose the model is correctly specified, such that X has pdf $f(\cdot,\theta_0)$ for some $\theta_0 \in \Theta$.

Fact. Then $\ell(\theta)$ is maximised at θ_0. This justifies using the MLE to approximate θ_0 when n is large.

Indeed,

$$\begin{aligned}
\ell(\theta) - \ell(\theta_0) &= \mathbb{E}_{\theta_0} \left(\log \frac{f(X,\theta)}{f(X,\theta_0)} \right) \\
&\le \log \mathbb{E}_{\theta_0} \left(\frac{f(X,\theta)}{f(X,\theta_0)} \right) \\
&= \log \int \frac{f(x,\theta)}{f(x,\theta_0)} f(x,\theta_0)\, dx \\
&= \log 1 = 0.
\end{aligned}$$

By Jensen's inequality, as log is concave, with equality if

$$\frac{f(X,\theta)}{f(X,\theta_0)}$$

is a.s. constant.

$$\therefore \quad \ell(\theta) \le \ell(\theta_0).$$

If

$$f(\cdot,\theta) = f(\cdot,\theta') \quad \Longrightarrow \quad \theta = \theta',$$

we say the parameter is identifiable.

In that case, if $\theta \neq \theta_0$, Jensen's inequality is strict, so θ_0 is actually the unique maximizer of ℓ.

Rmk. The difference

$$\ell(\theta_0) - \ell(\theta) = \int \log \frac{f(x,\theta_0)}{f(x,\theta)} f(x,\theta_0)\, dx$$

is called the Kullback–Leibler divergence or relative entropy between P_{θ_0} and P_θ.

A notion of distance which is not symmetric.

Obs 2.

$$0 = \frac{\partial}{\partial\theta} \underbrace{\int f(x,\theta)\,dx}_{=1\ \forall\theta} = \int \frac{\partial}{\partial\theta} f(x,\theta)\,dx$$
$$= \int \frac{\partial \log f(x,\theta)}{\partial\theta} f(x,\theta)\,dx$$
$$= \mathbb{E}_\theta\left[\frac{\partial \log f(X,\theta)}{\partial\theta}\right].$$

[There are some conditions for this. (Leibniz rule). See P & M.]
"$S_n(\theta)$ is centred when $X \sim P_\theta$."

Obs 3.

Def 5. Let $\Theta \subseteq \mathbb{R}^d$. For any $\theta \in \text{int}(\Theta)$, the Fisher Information matrix is

$$I(\theta) = \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X,\theta)\cdot\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^T\right], \qquad X \sim P_\theta.$$

Rmk. This is the covariance matrix of

$$S_n(\theta) = \frac{\partial}{\partial\theta}\log f(X,\theta)$$

when $X \sim P_\theta$.

Prop 1. For all $\theta \in \text{int}(\Theta)$, in a "regular" model,

$$I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(X,\theta)\right]$$

$$\left(I_{ij}(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta_i\,\partial\theta_j}\log f(X,\theta)\right]\right).$$

Proof.

$$0 = \frac{\partial^2}{\partial\theta\,\partial\theta^T}\int f(x,\theta)\,dx = \int \frac{\partial^2}{\partial\theta\,\partial\theta^T} f(x,\theta)\,dx.$$

[When this equality holds, we say the model is "regular".]
Now, we have

$$\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(x,\theta) = \frac{\partial}{\partial\theta^T}\left[\frac{1}{f(x,\theta)}\frac{\partial}{\partial\theta}f(x,\theta)\right]$$
$$= -\frac{1}{f(x,\theta)^2}\frac{\partial}{\partial\theta}f(x,\theta)\left(\frac{\partial}{\partial\theta}f(x,\theta)\right)^T + \frac{1}{f(x,\theta)}\frac{\partial^2}{\partial\theta\,\partial\theta^T}f(x,\theta).$$

Take $-\mathbb{E}_\theta[\cdot]$ of both sides.

$$-\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(X,\theta)\right]$$
$$= \mathbb{E}_\theta\left[\frac{1}{f(X,\theta)^2}\frac{\partial}{\partial\theta}f(X,\theta)\left(\frac{\partial}{\partial\theta}f(X,\theta)\right)^T\right] + 0$$
$$= \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X,\theta)\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^T\right] = I(\theta).$$

□

Tensorisation.

Suppose the observation X consists of n random variables $X_1, \ldots, X_n$. Then, the F.I. matrix is

$$I_n(\theta) = \mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n, \theta) \left(\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n, \theta) \right)^T \right].$$

If $X_1, \ldots, X_n$ are iid such that

$$f(X_1, \ldots, X_n, \theta) = \prod_{i=1}^{n} f(X_i, \theta),$$

where $\{f(x, \theta) : \theta \in \Theta\}$ is a model for X_1, then the F.I. is

$$\begin{aligned} I_n(\theta) &= \mathbb{E}_\theta \left[\sum_{i,j} \underbrace{\frac{\partial}{\partial \theta} \log f(X_i, \theta)}_{\substack{\text{mean 0 under} \\ P_\theta}} \left(\frac{\partial}{\partial \theta} \log f(X_j, \theta) \right)^T \right] \\ &= \mathbb{E}_\theta \left[\sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(X_i, \theta) \left(\frac{\partial}{\partial \theta} \log f(X_i, \theta) \right)^T \right] \\ &= nI(\theta), \end{aligned}$$

where $I(\theta)$ is the Fisher Information for a single observation.

End of lecture 2

Recap of Information Geometry

Observation 1.

$$\mathrm{KL}(P_{\theta_0}, P_\theta) = \mathbb{E}_{\theta_0} \left[\log \frac{f(X, \theta_0)}{f(X, \theta)} \right] \geq 0,$$

minimised at $\theta = \theta_0$.

Observation 2.

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X, \theta) \right] = 0, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Observation 3.

$$I(\theta) := \mathrm{Var}_\theta \left[\frac{\partial}{\partial \theta} \log f(X, \theta) \right] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta \, \partial \theta^T} \log f(X, \theta) \right], \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Why geometry? The F.I., $I(\theta)$, defines a metric tensor on the manifold

$$\{f(\cdot, \theta) : \theta \in \Theta\}.$$

This is known as the **Fisher–Rao metric**. KL divergence is approximated locally by this metric.

- **Reparametrisation invariance.** Take the Bernoulli(θ) model, where θ is the probability of success. Reparametrise it in terms of "log-odds":

$$\eta = \log\left(\frac{\theta}{1-\theta}\right) \in \mathbb{R}.$$

  The manifold

$$\{f(\cdot, \eta) : \eta \in \mathbb{R}\},$$

  equipped with the metric tensor $I(\eta)$, preserves geodesics, path lengths, etc.

- **Sufficient statistic invariance.**

Chentsov's theorems. These properties characterise the Fisher–Rao metric.

Proposition 2 (Cramér–Rao Lower Bound)

Let

$$\{f(\cdot, \theta) : \theta \in \Theta \subseteq \mathbb{R}\}$$

be a "regular" statistical model. Let $\tilde{\theta}(X_1, \ldots, X_n)$ be an unbiased estimator based on i.i.d. samples $X_1, \ldots, X_n$:

$$\mathbb{E}_\theta \tilde{\theta} = \theta, \qquad \forall \theta \in \Theta.$$

Then

$$\mathrm{Var}_\theta(\tilde{\theta}) \geq \frac{1}{nI(\theta)}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Proof. Recall the Cauchy–Schwarz inequality for random variables Y, Z:

$$\big(\mathbb{E}[YZ]\big)^2 \leq \mathbb{E}[Y^2]\mathbb{E}[Z^2].$$

Let

$$Z = \tilde{\theta} - \theta, \qquad Y = \frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta).$$

Then

$$\mathbb{E}[Z^2] = \mathrm{Var}_\theta(\tilde{\theta}), \qquad \text{as } \tilde{\theta} \text{ is unbiased,}$$

and

$$\mathbb{E}[Y^2] = nI(\theta), \qquad \text{as the F.I. tensorises.}$$

Moreover,

$$\begin{aligned}
\mathbb{E}[YZ] &= \mathbb{E}_\theta\left[\tilde{\theta}\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta)\right] \\
&\quad - \theta \underbrace{\mathbb{E}_\theta\left[\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta)\right]}_{=0} \\
&= \int_{\mathcal{X}^n} \tilde{\theta}(x_1, \ldots, x_n) \frac{\partial f(x_1, \ldots, x_n; \theta)}{\partial \theta} \frac{f(x_1, \ldots, x_n; \theta)}{f(x_1, \ldots, x_n; \theta)}\, dx_1 \cdots dx_n \\
&= \frac{\partial}{\partial \theta} \underbrace{\int_{\mathcal{X}^n} \tilde{\theta}\, f(x_1, \ldots, x_n; \theta)\, dx_1 \cdots dx_n}_{=\mathbb{E}_\theta \tilde{\theta} = \theta} \qquad \text{(by regularity)} \\
&= \frac{\partial \theta}{\partial \theta} = 1.
\end{aligned}$$

Plug these expectations into the Cauchy–Schwarz inequality and rearrange. □

Remark. Sometimes, the MLE is unbiased and attains the Cramér–Rao lower bound, but not always. However, we will show this is true asymptotically.

If $\Theta \subseteq \mathbb{R}^p$ with $p > 1$, let

$$\Phi : \Theta \to \mathbb{R}$$

be a scalar function of θ. Consider an unbiased estimator $\tilde{\Phi}(X_1, \dots, X_n)$. Then one can show that

$$\mathrm{Var}_\theta(\tilde{\Phi}) \geq \frac{1}{n} \left(\frac{\partial \Phi(\theta)}{\partial \theta} \right)^T I(\theta)^{-1} \frac{\partial \Phi(\theta)}{\partial \theta}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Corollary 1. If $\tilde{\theta}$ is not unbiased, we have

$$\mathrm{Var}_\theta(\tilde{\theta}) \geq \frac{\dfrac{\partial}{\partial \theta} \mathbb{E}_\theta \tilde{\theta}}{nI(\theta)}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Example. Take $\Phi(\theta) = \alpha^T \theta$ for some $\alpha \in \mathbb{R}^p$. In this case,

$$\frac{\partial \Phi(\theta)}{\partial \theta} = \alpha,$$

and

$$\mathrm{Var}_\theta(\tilde{\Phi}) \geq \frac{1}{n} \alpha^T I(\theta)^{-1} \alpha.$$

Example Sheet 1

We observe a single pair

$$(X_1, X_2) \sim N\left(\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \Sigma \right),$$

where $\Sigma \succ 0$ is known. We wish to estimate θ_1.

Case 1. Assume that θ_2 is known. In this case, the model has one parameter and F.I. matrix

$$I_1(\theta_1).$$

Case 2. Assume that θ_2 is unknown. For any estimator

$$\tilde{\theta} = \begin{pmatrix} \tilde{\theta}_1 \\ \tilde{\theta}_2 \end{pmatrix},$$

we have

$$\tilde{\theta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \tilde{\theta}.$$

So, if $\tilde{\theta}$ is unbiased, we have the Cramér–Rao lower bound

$$\mathrm{Var}_\theta(\tilde{\theta}_1) \geq \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T I(\theta)^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left[I(\theta)^{-1} \right]_{11}.$$

Goal. Show

$$I_1(\theta_1)^{-1} < \left[I(\theta)^{-1} \right]_{11},$$

unless Σ is diagonal (X_1, X_2 independent).

II. Asymptotic Theory of the MLE

All reasonable estimators are asymptotically unbiased:

$$\mathbb{E}_\theta \tilde{\theta} \xrightarrow{n\to\infty} \theta, \qquad \forall \theta \in \Theta.$$

We wish to prove that estimators are **consistent**:

$$\tilde{\theta} \xrightarrow{?} \theta \qquad \text{as } n \to \infty \quad \text{under } P_\theta, \qquad \forall \theta \in \Theta.$$

For consistent estimators, a reasonable notion of optimality is **asymptotic efficiency**:

$$n \operatorname{Var}_\theta(\tilde{\theta}) \xrightarrow{n\to\infty} I(\theta)^{-1}, \qquad \forall \theta \in \operatorname{int}(\Theta).$$

We will show that the MLE is asymptotically efficient and, furthermore,

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, n^{-1} I(\theta)^{-1}\right), \qquad \forall \theta.$$

This distributional convergence can be used for statistical inference.

Stochastic Convergence

Definition 1. Let $(X_n : n \in \mathbb{N})$ be a collection of random variables in some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- We say that X_n converges almost surely to a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$, or

  $$X_n \xrightarrow{\text{a.s.}} X,$$

  if and only if

  $$\mathbb{P}\left(\left\{\omega : \|X_n(\omega) - X(\omega)\|_2 \xrightarrow{n\to\infty} 0\right\}\right) = 1.$$

- We say that X_n converges to X in probability, or

  $$X_n \xrightarrow{P} X,$$

  if, for every $\varepsilon > 0$,

  $$\mathbb{P}\left(\|X_n - X\|_2 > \varepsilon\right) \xrightarrow{n\to\infty} 0.$$

End of lecture 3

Recall: If $(X_n), X$ are random vectors in $\mathbb{R}^k$, we say

- $X_n \xrightarrow{\text{a.s.}} X$ iff

  $$\mathbb{P}\left(\|X_n - X\|_2 \xrightarrow{n\to\infty} 0\right) = 1.$$

- $X_n \xrightarrow{P} X$ iff

  $$\forall \varepsilon > 0, \qquad \mathbb{P}\left(\|X_n - X\|_2 > \varepsilon\right) \xrightarrow{n\to\infty} 0.$$

Rmk.

$$X_n \xrightarrow{\text{a.s./}P} X \quad \text{iff} \quad X_{n,i} \xrightarrow{\text{a.s./}P} X_i \qquad \forall\, 1 \le i \le k.$$

$$[X \le t \iff X_i \le t_i \quad \forall\, 1 \le i \le k]$$

Def 2. We say that X_n converges in distribution to X, or $X_n \xrightarrow{d} X$, if for all $t \in \mathbb{R}^k$ where the function

$$t \mapsto \mathbb{P}(X \leq t)$$

is continuous, we have

$$\mathbb{P}(X_n \leq t) \xrightarrow{n\to\infty} \mathbb{P}(X \leq t).$$

Some facts from P & M.

Prop 1.

(a)

$$X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X.$$

(b) Continuous Mapping Theorem (CMT). Let $(X_n), X$ take values in $\mathcal{X} \subseteq \mathbb{R}^k$, let $g : \mathcal{X} \to \mathbb{R}$ be continuous. Then,

$$X_n \xrightarrow{\text{a.s.}/P/d} X \implies g(X_n) \xrightarrow{\text{a.s.}/P/d} g(X).$$

(c) Slutsky's Lemma. Suppose that

$$X_n \xrightarrow{d} X, \qquad Y_n \xrightarrow{d} c,$$

where c is constant, then

(i)

$$Y_n \xrightarrow{P} c.$$

(ii)

$$X_n + Y_n \xrightarrow{d} X + c.$$

(iii) If $k = 1$,

$$X_n Y_n \xrightarrow{d} X \cdot c,$$
$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c} \qquad \text{if } c \neq 0.$$

(iv) Let (A_n) be a sequence of random matrices in $\mathbb{R}^{m\times k}$, and suppose,

$$(A_n)_{ij} \xrightarrow{P} A_{ij} \qquad \forall i, j$$

where A is deterministic. Then,

$$A_n X_n \xrightarrow{d} AX.$$

(d) If $X_n \xrightarrow{d} X$, then (X_n) is "bounded in probability" or $X_n = O_P(1)$, i.e. for all $\varepsilon > 0$, $\exists\, M_\varepsilon > 0$, $N_\varepsilon \in \mathbb{N}$ s.t.

$$\mathbb{P}\left(\|X_n\|_2 > M_\varepsilon\right) < \varepsilon \qquad \forall n \geq N_\varepsilon.$$

2. LLN & CLT

Let $X_1, X_2, \ldots$ be iid copies of a r.v. X in $\mathbb{R}^k$. When $k = 1$, the Weak Law of large numbers says that if $\text{Var}(X) < \infty$, then

$$\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} \mathbb{E}X.$$

Indeed by Chebyshev's inequality,

$$\forall \varepsilon > 0, \qquad \mathbb{P}\left(|\bar{X}_n - \mathbb{E}X| > \varepsilon\right) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\mathrm{Var}(X)}{n\varepsilon^2} \xrightarrow{n\to\infty} 0.$$

Theorem 1 (Strong law of large numbers). If $\mathbb{E}\left(\|X_n\|_2\right) < \infty$, then

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}[X] \qquad \text{as } n \to \infty.$$

Theorem 2 (Central Limit Theorem). When $k = 1$, assume in addition that $\mathrm{Var}(X) = \sigma^2 < \infty$, then,

$$\sqrt{n}\left(\bar{X}_n - \mathbb{E}X\right) \xrightarrow{d} N(0, \sigma^2).$$

To state a multivariate version, recall that $Z \sim N(\mu, \Sigma)$ iff

$$\forall \alpha \in \mathbb{R}^k, \qquad \alpha^T Z \sim N(\alpha^T \mu, \alpha^T \Sigma \alpha).$$

If Σ is positive definite, then Z has a p.d.f.

$$f(z) = \frac{1}{(2\pi)^{n/2}} \frac{1}{|\det \Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\right\}, \qquad z \in \mathbb{R}^k.$$

Theorem 2 ($k \geq 1$). If $\mathrm{Var}(X) = \Sigma$ is finite & pos. def. then

$$\sqrt{n}\left(\bar{X}_n - \mathbb{E}X\right) \xrightarrow{d} N(0, \Sigma).$$

Cor. By Prop 1(d),

$$\sqrt{n}\left(\bar{X}_n - \mathbb{E}X\right) = O_P(1) \iff \left(\bar{X}_n - \mathbb{E}X\right) = O_P\left(\frac{1}{\sqrt{n}}\right).$$

The CLT is useful to define confidence intervals. Let $X_1, X_2, \ldots$ be iid copies of X with mean μ_0 and $\mathrm{Var}(X) = \sigma^2 < \infty$ known. Define

$$C_n = \left\{\mu \in \mathbb{R} : \sqrt{n}\frac{|\bar{X}_n - \mu|}{\sigma} \leq z_\alpha\right\}$$

where z_α is such that

$$\mathbb{P}(|Z| \leq z_\alpha) = 1 - \alpha \qquad \text{if } Z \sim N(0,1).$$

$$\begin{aligned}
\mathbb{P}(\mu_0 \in C_n) &= \mathbb{P}\left(\sqrt{n}|\bar{X}_n - \mu_0| \leq \sigma z_\alpha\right) \\
&\xrightarrow{n\to\infty} \mathbb{P}(|Z| \leq z_\alpha) \qquad \text{by the CLT} \\
&= 1 - \alpha.
\end{aligned}$$

Hence, we say that C_n is an "asymptotic" $(1-\alpha)$-confidence interval.

If σ^2 is unknown, we can define an estimator

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n-1}\sum_{i=1}^n \left(X_i - \underbrace{\bar{X}_n}_{\frac{1}{n}\sum_j X_j}\right)^2 \\
&= \frac{1}{n-1}\left(\sum_{i=1}^n X_i^2 - n\bar{X}_n^2\right) \\
&\xrightarrow{d} \mathbb{E}(X^2) - (\mathbb{E}X)^2 = \sigma^2
\end{aligned}$$

by several applications of the SLLN, the CMT and Slutsky's lemma.

Define a new random interval

$$\widetilde{C}_n = \left\{ \mu \in \mathbb{R} : \sqrt{n} \frac{|\bar{X}_n - \mu|}{\hat{\sigma}} \leq z_\alpha \right\}.$$

By the CMT, $\hat{\sigma} \xrightarrow{d} \sigma$ constant. By Slutsky's Lemma,

$$\sqrt{n} \frac{(\bar{X}_n - \mu_0)}{\hat{\sigma}} \xrightarrow{d} N(0,1).$$

Hence, C_n is an asymptotic $(1-\alpha)$-confidence interval for μ_0.

Next time: Uniform laws of Large Numbers (ULLN)

Let $\mathcal{H}$ be a class of functions $\mathbb{R}^k \to \mathbb{R}$. Assume that

$$\mathbb{E}|h(X)| < \infty \qquad \forall h \in \mathcal{H}.$$

We know from the SLLN that

$$\frac{1}{n} \sum_{i=1}^{n} h(X_i) \xrightarrow{\text{a.s.}} \mathbb{E}[h(X)] \qquad \forall h \in \mathcal{H}.$$

We have a ULLN if

$$\sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^{n} h(X_i) - \mathbb{E}[h(X)] \right| \xrightarrow{\text{a.s.}} 0.$$

End of lecture 4

Uniform Laws of Large Numbers

Let X be a r.v. on $\mathcal{X}$. (X_i) iid copies of X. Let $\mathcal{H}$ be a class of functions $\mathcal{X} \to \mathbb{R}$, s.t.

$$\mathbb{E}|h(X)| < \infty \qquad \forall h \in \mathcal{H}.$$

$$\text{ULLN} \qquad \sup_{h \in \mathcal{H}} \left| \underbrace{\frac{1}{n} \sum_{i=1}^{n} h(X_i)}_{=P_n h} - \underbrace{\mathbb{E}h(X)}_{=Ph} \right| \xrightarrow{\text{a.s.}} 0.$$

If $\mathcal{H} = \{h_1, \ldots, h_M\}$ is finite, then by the SLLN, for any $1 \leq j \leq M$,

$$|P_n h_j - P h_j| \xrightarrow{\text{a.s.}} 0$$

i.e. on an event A_j with $\mathbb{P}(A_j) = 1$, $\forall \varepsilon > 0$, $\exists N_j(\varepsilon)$ s.t.

$$|P_n h_j - P h_j| < \varepsilon \qquad \forall n > N_j(\varepsilon).$$

Hence on $A := \bigcap_{j=1}^{M} A_j$, we have $\forall \varepsilon > 0$,

$$\max_{1 \leq j \leq M} |P_n h_j - P h_j| < \varepsilon \qquad \forall n > \max_j N_j(\varepsilon).$$

As $\mathbb{P}(A) = 1$, this means

$$\sup_{h \in \mathcal{H}} |P_n h - P h| \xrightarrow{\text{a.s.}} 0.$$

When $|\mathcal{H}|$ is infinite, we will use a "bracketing argument". A bracket is a collection of functions $\{\underline{h}_j, \overline{h}_j\}_{j=1}^{M}$ such that

$$\forall j, \qquad \underline{h}_j(x) \leq \overline{h}_j(x) \qquad \forall x \in \mathcal{X}.$$

A finite bracket covers $\mathcal{H}$ if for all $h \in \mathcal{H}$, $\exists j \in \{1, \ldots, M\}$ s.t.

$$\underline{h}_j(x) \leq h(x) \leq \overline{h}_j(x) \qquad \forall x \in \mathcal{X}.$$

Prop 2. Suppose that $\forall \varepsilon > 0$, there is a finite bracket $\{\underline{h}_j, \overline{h}_j\}_{j=1}^M$ covering $\mathcal{H}$, s.t.

$$(1) \quad \mathbb{E}|\underline{h}_j(X)| < \infty, \quad \mathbb{E}|\overline{h}_j(X)| < \infty \qquad \forall 1 \le j \le M,$$
$$(2) \quad \mathbb{E}|\overline{h}_j(X) - \underline{h}_j(X)| < \varepsilon.$$

Then,

$$\sup_{h \in \mathcal{H}} |P_n h - Ph| \xrightarrow{\text{a.s.}} 0.$$

Proof. Take any $\varepsilon > 0$, then for any $h \in \mathcal{H}$, $\exists 1 \le j \le M$ s.t. $\underline{h}_j \le h \le \overline{h}_j$. Then

$$\begin{aligned} P_n h - Ph &\le P_n \overline{h}_j - P\underline{h}_j \\ &\le |P_n \overline{h}_j - P\overline{h}_j| + \underbrace{|P(\overline{h}_j - \underline{h}_j)|}_{\le \varepsilon}. \end{aligned}$$

Similarly,

$$Ph - P_n h \le P\overline{h}_j - P_n \underline{h}_j \le |P_n \underline{h}_j - P\underline{h}_j| + \underbrace{|P(\overline{h}_j - \underline{h}_j)|}_{\le \varepsilon}.$$

$$\therefore \quad \sup_{h \in \mathcal{H}} |P_n h - Ph| \le \underbrace{\max_{g \in \text{bracket}} |P_n g - Pg|}_{\xrightarrow{\text{a.s.}} 0 \text{ by the SLLN as bracket is finite}} + \varepsilon.$$

Hence, on an event A_ε with $\mathbb{P}(A_\varepsilon) = 1$, we have

$$\limsup_{n \to \infty} \sup_{h \in \mathcal{H}} |P_n h - Ph| \le \varepsilon.$$

Take sequence $\varepsilon_k \downarrow 0$. Let $A = \bigcap_{k=1}^\infty A_{\varepsilon_k}$, with $\mathbb{P}(A) = 1$. On A,

$$\sup_{h \in \mathcal{H}} |P_n h - Ph| \longrightarrow 0. \qquad \square$$

3 Consistency of the MLE

Def 3. We say an estimator $\hat{\theta}(X_1, \ldots, X_n)$ is consistent if $\hat{\theta} \xrightarrow{P} \theta$ under P_θ $\forall \theta \in \Theta$.

Assumptions. Let $\{f(\cdot, \theta) : \theta \in \Theta\}$ be a model of pdfs or pmfs on $\mathcal{X} \subseteq \mathbb{R}^d$. We'll assume

(1) $f(x, \theta) > 0 \qquad \forall x \in \mathcal{X},\ \theta \in \Theta$.

(2) $\theta \mapsto f(x, \theta)$ is continuous $\forall x \in \mathcal{X}$.

(3) Θ is compact.

(4) Strict identifiability: $f(\cdot, \theta) = f(\cdot, \theta') \implies \theta = \theta'$.

(5) $\mathbb{E}_\theta \underbrace{\sup_{\theta' \in \Theta} |\log f(X, \theta')|}_{M(X)} < \infty \qquad \forall \theta \in \Theta$.

Recall that (5) allows us to define $\ell(\theta) := \mathbb{E}_{\theta_0} \log f(X, \theta)$ for a true parameter θ_0. Furthermore, by (4) $\ell(\theta)$ is maximised uniquely at $\theta = \theta_0$.

By (2) & (5), we have that ℓ is continuous. Indeed take (θ_i), $\theta_i \to \theta$, then

$$|\ell(\theta_i) - \ell(\theta)| = \left| \mathbb{E}_{\theta_0} \left[\underbrace{\log f(X, \theta_i) - \log f(X, \theta)}_{\xrightarrow{\text{a.s.}} 0 \text{ by (2)}} \right] \right|$$
$$\longrightarrow 0 \quad \text{by the Dominated Convergence Theorem (P\&M) using the domination condition (5).}$$

In maximum likelihood we maximise

$$\bar{\ell}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i, \theta), \qquad \text{a "sample" version of } \ell(\theta).$$

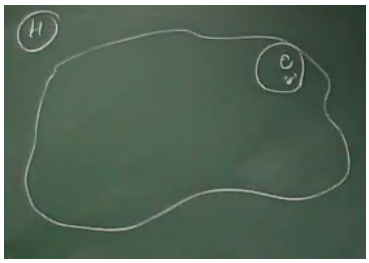
Lemma. If (X_i) are iid with law P_{θ_0}, we have the ULLN

$$\sup_{\theta\in\Theta} \left| \underbrace{\bar{\ell}_n(\theta)}_{=P_n h_\theta} - \underbrace{\ell(\theta)}_{=Ph_\theta} \right| \xrightarrow{\text{a.s.}} 0$$

where $h_\theta : x \mapsto \log f(x, \theta)$. Here $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$.

Proof sketch To apply Prop 2, we need to construct for any $\varepsilon > 0$ a finite bracket satisfying the hypotheses (1) & (2). Take any $\theta \in \Theta$, let $B_{\theta,\delta} = \{\theta' : \|\theta - \theta'\| \leq \delta\}$, let

$$\overline{h}_\theta^{(\delta)} = \sup_{\theta' \in B_{\theta,\delta}} h_{\theta'}(x), \qquad \underline{h}_\theta(x) = \inf_{\theta' \in B_{\delta,\theta}} h_{\theta'}(x).$$



By def

$$\underline{h}_\theta \leq h_{\theta'} \leq \overline{h}_\theta \qquad \forall \theta' \in B_{\delta,\theta}.$$

Hypothesis (1) is given by Assumption (5). We'll choose δ small enough so that we can obtain Hypothesis 2:

$$\mathbb{E}|\overline{h}_\theta(X) - \underline{h}_\theta(X)| < \varepsilon$$

(To do this carefully, apply the DCT using assumptions (2) & (5).)

For each $\theta \in \Theta$ we obtain a pair $(\underline{h}_\theta, \overline{h}_\theta)$ sandwiching functions in a "ball" around h_θ. By compactness, $\exists$ a finite bracket $\{\overline{h}_{\theta_j}, \underline{h}_{\theta_j}\}_{j=1}^{M}$ s.t. $\forall \theta \in \Theta$, $\exists j$ s.t.

$$\underline{h}_{\theta_j} \leq h_\theta \leq \overline{h}_{\theta_j}.$$

This allows us to apply prop 2. $\square$

End of lecture 5

Recall. We stated 5 assumptions for consistency of the MLE.
Implications

- $\ell(\theta) = \mathbb{E}_{\theta_0}[\log f(X, \theta)] < \infty \ \forall \theta$, maximised uniquely at θ_0.
- ℓ is continuous on Θ.
- ULLN

$$\sup_{\theta\in\Theta} |\bar{\ell}_n(\theta) - \ell(\theta)| \xrightarrow{\text{a.s./}P/d} 0.$$

Theorem 3. Suppose that $\{f(\cdot,\theta) : \theta \in \Theta\}$ satisfies the stated assumptions and $X_1, X_2, \ldots$ are iid P_{θ_0}. Then an MLE $\hat{\theta}_n$ based on $X_1, \ldots, X_n$ exists and

$$\hat{\theta}_n \xrightarrow{P} \theta_0.$$

Proof.

$$\bar{\ell}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i, \theta)$$

is continuous, so it attains a supremum in Θ compact at some $\hat{\theta}_n \in \Theta$.

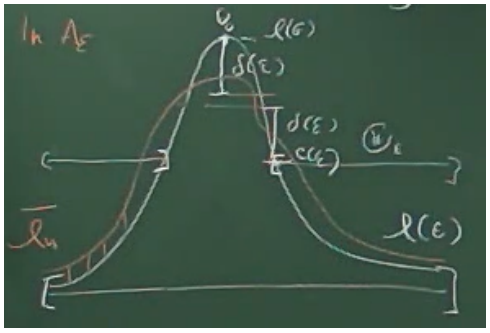
Define

$$\Theta_\varepsilon = \{\theta \in \Theta : \|\theta - \theta_0\|_2 \geq \varepsilon\}$$

and note Θ_ε is compact (we remove an open ball from Θ compact). We'll show that

$$\mathbb{P}(\hat{\theta}_n \in \Theta_\varepsilon) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

If this holds for arbitrary $\varepsilon > 0$, $\hat{\theta}_n \xrightarrow{P} \theta_0$.



$$\sup_{\theta \in \Theta_\varepsilon} \ell(\theta) =: c(\varepsilon) < \ell(\theta_0).$$

Choose

$$\delta(\varepsilon) < \frac{\ell(\theta_0) - c(\varepsilon)}{2}.$$

Define an event

$$A_\varepsilon = \left\{ \sup_{\theta \in \Theta} |\bar{\ell}_n(\theta) - \ell(\theta)| \leq \delta(\varepsilon) \right\}.$$

By the ULLN, $\mathbb{P}(A_\varepsilon) \to 1$ as $n \to \infty$.

$$\begin{aligned}
\sup_{\theta \in \Theta_\varepsilon} \bar{\ell}_n(\theta) &\leq \sup_{\theta \in \Theta_\varepsilon} \ell(\theta) + \underbrace{\sup_{\theta \in \Theta_\varepsilon} (\bar{\ell}_n(\theta) - \ell(\theta))}_{\leq \delta(\varepsilon) \text{ on } A_\varepsilon} \\
&\leq c(\varepsilon) + \delta(\varepsilon) \\
&< \ell(\theta_0) - \delta(\varepsilon) \qquad \text{(by def of } \delta(\varepsilon)) \\
&\leq \bar{\ell}_n(\theta_0).
\end{aligned}$$

Hence, on A_ε, $\hat{\theta}_n \notin \Theta_\varepsilon$. In other words $A_\varepsilon \subseteq \{\hat{\theta}_n \notin \Theta_\varepsilon\}$. Therefore

$$\mathbb{P}(\hat{\theta}_n \notin \Theta_\varepsilon) \geq \mathbb{P}(A_\varepsilon) \longrightarrow 1 \qquad \text{as } n \to \infty.$$

Finally

$$\mathbb{P}(\hat{\theta}_n \in \Theta_\varepsilon) \longrightarrow 0 \qquad \text{as } n \to \infty. \qquad \square$$

Rmk. If we assume that $\theta \mapsto \log f(x,\theta)$ is differentiable for all $x \in \mathcal{X}$, and $\frac{\partial \bar{\ell}_n(\theta)}{\partial \theta}$ satisfies a ULLN and its limit $\frac{\partial \ell(\theta)}{\partial \theta}$ has a single 0 at θ_0, there is a simpler argument to show consistency. Can deal with noncompact Θ in some simple models.

4. Asymptotic Normality of the MLE

Fluctuations of $\sqrt{n}(\hat{\theta}_n - \theta_0)$.

Theorem 4. Suppose that $\{f(\cdot,\theta) : \theta \in \Theta\}$ satisfies the 9 stated assumptions. Let $\hat{\theta}_n$ be an MLE based on $X_1, \ldots, X_n$ iid P_{θ_0}. Then as $n \to \infty$,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)).$$

The nine assumptions are the five assumptions for consistency stated above, together with the following four additional regularity assumptions (R2).

Regularity assumptions (R2). Let the statistical model be such that:

(1) $\theta_0 \in \operatorname{int} \Theta$;

(2) there exists an open neighbourhood N of θ_0 on which $\theta \mapsto f(x,\theta)$ is twice continuously differentiable for all $x \in \mathcal{X}$;

(3) $I(\theta_0)$ exists and is invertible;

(4)
$$\int_{\mathcal{X}} \sup_{\theta \in \Theta} \|\nabla_\theta \log f(x,\theta)\|_2 \, dx < \infty$$
$$\mathbb{E}_{\theta_0}\left(\sup_{\theta \in \Theta} \|\nabla^2_\theta \log f(X,\theta)\|_2 \right) < \infty$$
$$\int_{\mathcal{X}} \sup_{\theta \in \Theta} \|\nabla^2_\theta \log f(x,\theta)\|_2 \, dx < \infty.$$

(In fact the above can be weakened by replacing Θ above with any compact set K with non-empty interior that contains θ_0.)

End of lecture 6

Theorem 4 (Asymp. Normality of MLE). Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_{\theta_0}$, and the 9 stated assumptions hold. Then the MLE $\hat{\theta}_n$ satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}).$$

Proof. Recall that

$$\ell_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log f(X_i, \theta).$$

Step 1. Taylor expansion. Choose a compact neighbourhood $K \subset \operatorname{int}\Theta$ of θ_0 on which the regularity conditions hold, and let

$$E_n = \{\|\hat{\theta}_n - \theta_0\| < \varepsilon\},$$

where $\varepsilon > 0$ is sufficiently small that the line segment $[\theta_0, \hat{\theta}_n]$ lies in K on E_n. By consistency of MLE,

$$\mathbb{P}(E_n) \to 1.$$

On E_n, $\|\hat{\theta}_n - \theta_0\| < \varepsilon$, so $B_\varepsilon(\theta_0) \subseteq \operatorname{int}(K) \subset \operatorname{int}(\Theta)$ given ε small. Hence, the MLE is the interior maximiser. So

$$\nabla \ell_n(\hat{\theta}_n) = 0.$$

Apply MVT componentwise:

$$\nabla \ell_n(\hat{\theta}_n) = 0 = \nabla \ell_n(\theta_0) + A_n(\hat{\theta}_n - \theta_0),$$

where

$$(A_n)_{jk} = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta_j \partial\theta_k} \log\big(f(X_i, \theta^{(j)})\big), \qquad \theta^{(j)} \in [\theta_0, \hat{\theta}_n].$$

Step 2. Prove $A_n \xrightarrow{P} -I(\theta_0)$. Fix j, k, define

$$g(x,\theta) = \frac{\partial^2}{\partial\theta_j \partial\theta_k} \log f(x,\theta), \qquad \bar{g}(\theta) = \mathbb{E}_{\theta_0}[g(X,\theta)].$$

By equivalent form of Fisher information,

$$\bar{g}(\theta_0) = -I(\theta_0)_{jk}.$$

Hence,

$$\begin{aligned} |(A_n)_{jk} + I(\theta_0)_{jk}| &= \left| \frac{1}{n}\sum_{i=1}^{n} g(X_i, \theta^{(j)}) - \bar{g}(\theta_0) \right| \\ &\le \underbrace{\sup_{\theta \in K} \left| \frac{1}{n}\sum_{i=1}^{n} g(X_i,\theta) - \bar{g}(\theta) \right|}_{\text{(I)}} + \underbrace{|\bar{g}(\theta^{(j)}) - \bar{g}(\theta_0)|}_{\text{(II)}}. \end{aligned}$$

For (I), by continuity of $g(x,\cdot)$, compactness of K, and

$$\mathbb{E}_{\theta_0}\left[\sup_{\theta\in K} |g(X,\theta)|\right] < \infty,$$

and the "bracket" version of ULLN, we can in a similar way deduce that g has ULLN here, namely

$$\text{(I)} \xrightarrow{\text{a.s.}} 0.$$

For (II), since $\theta^{(j)} \in [\theta_0, \hat{\theta}_n]$,

$$\|\theta^{(j)} - \theta_0\| \le \|\hat{\theta}_n - \theta_0\| \xrightarrow{P} 0$$

by consistency of MLE. Moreover, DCT implies that $\bar{g}$ is cts. at θ_0, thus by CMT,

$$\text{(II)} \xrightarrow{P} 0.$$

Thus, for $\forall j, k$,

$$(A_n)_{jk} \xrightarrow{P} -I(\theta_0)_{jk}.$$

Result follows. □

Since $I(\theta_0)$ is invertible, by CMT on $\det(\cdot)$, A_n is invertible with probability tending to 1 (i.e., in probability).

Step 3. Apply multivariate CLT. Define

$$S_i = \nabla_\theta \log f(X_i, \theta_0).$$

By regularity conditions,

$$\mathbb{E}_{\theta_0}[S_i] = 0, \qquad \mathrm{Var}_{\theta_0}(S_i) = I(\theta_0).$$

By CLT,

$$\sqrt{n}\,\nabla \ell_n(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} S_i \xrightarrow{d} N(0, I(\theta_0)).$$

Step 4. Apply Slutsky's Lemma. On E_n, whenever A_n is invertible, Taylor expansion gives

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -A_n^{-1}\sqrt{n}\,\nabla \ell_n(\theta_0). \qquad (*)$$

By Slutsky's Lemma, and claim,

$$-A_n^{-1}\sqrt{n}\,\nabla \ell_n(\theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1} I(\theta_0) I(\theta_0)^{-1}).$$

(Using fact: $X_n \sim N(\mu, \Sigma) \Rightarrow AX_n \sim N(A\mu, A\Sigma A^T)$.) So

$$-A_n^{-1}\sqrt{n}\,\nabla \ell_n(\theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}).$$

Finally, since $\mathbb{P}(E_n) \to 1$ and $\mathbb{P}(A_n \text{ invertible}) \to 1$, $(*)$ holds with probability tending to 1. So the difference $\xrightarrow{P} 0$. So using Slutsky again,

$$X_n \xrightarrow{d} X, \quad Y_n \xrightarrow{P} 0 \quad \Rightarrow \quad X_n + Y_n \xrightarrow{d} X.$$

Result follows. □

(1) The continuous derivatives of $\theta \mapsto \log f(x, \theta)$ are <u>not</u> necessary for asymptotic normality. Le Cam theory relaxes this for models with a single weak derivative.

<u>Ex</u> Laplace model

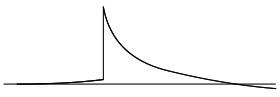
$$f(x, \theta) = \frac{1}{2}\exp\{-|x - \theta|\}, \qquad x \in \mathbb{R}.$$

(2) One weak derivative is necessary.

<u>Ex</u> $X \sim \mathrm{Unif}([0, \theta])$, $\theta > 0$.

The MLE is $\hat{\theta}_n = \max_{1 \le i \le n} X_i$, and

$$f(x, \theta) = \frac{1}{\theta}\,\mathbf{1}_{\{x \le \theta\}}.$$



The likelihood is discontinuous at $\theta = x$ and $\sqrt{n}(\hat{\theta}_n - \theta_0)$ has a non-normal limit (Ex sheet).

(3) It is also necessary that $\theta_0 \in \mathrm{int}(\Theta)$.

<u>Ex</u> $X \sim N(\theta, 1)$, where $\Theta = [0, \infty)$.

Here the MLE $\hat{\theta}_n = \max(0, \overline{X}_n)$. Under P_{θ_0} where $\theta_0 = 0$ (on the boundary of $[0, \infty)$), we can show that $\sqrt{n}(\hat{\theta}_n - \theta_0)$ has a non-normal limit (Ex sheet).

(4) Cramer Rao lower bound says that $I^{-1}(\theta_0)$ is best variance for $\sqrt{n}(\hat{\theta}_n - \theta_0)$ for unbiased $\hat{\theta}_n$, for any finite n.

Le Cam theory In models with the "local asymptotic normality", any estimator $\tilde{\theta}_n$ which is "regular" has asymptotic variance $\geq I^{-1}(\theta_0)$. Non-regular estimators can have better asymptotic variance for certain values of θ_0.

Ex Hodge's estimator

Let $\hat{\theta}_n$ be the MLE in a regular model s.t.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)) \qquad \forall \theta_0 \in \text{int}(\Theta).$$

Define

$$\tilde{\theta}_n = \begin{cases} 0 & \text{if } |\hat{\theta}_n| < n^{-1/4}, \\ \hat{\theta}_n & \text{o.w.} \end{cases}$$

If $\theta_0 \neq 0$, under P_{θ_0},

$$\begin{aligned} \mathbb{P}(\tilde{\theta}_n \neq \hat{\theta}_n) &= \mathbb{P}\big(\underbrace{|\hat{\theta}_n|}_{|\theta_0| - |\hat{\theta}_n - \theta_0| \leq} < n^{-1/4} \big) \\ &\leq \mathbb{P}\big(|\theta_0| - |\hat{\theta}_n - \theta_0| < n^{-1/4}\big) \\ &= \mathbb{P}\big(|\hat{\theta}_n - \theta_0| > |\theta_0| - n^{-1/4}\big) \xrightarrow{n\to\infty} 0 \qquad \text{as } \hat{\theta}_n \text{ consistent.} \end{aligned}$$

Therefore, under P_{θ_0}, $\sqrt{n}(\tilde{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0))$. However, if $\theta_0 = 0$,

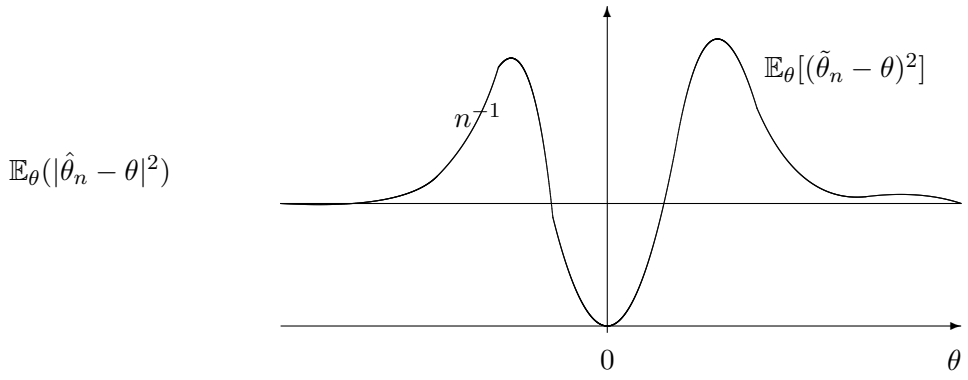
$$\begin{aligned} \mathbb{P}(\hat{\theta}_n = \tilde{\theta}_n) &= \mathbb{P}(|\hat{\theta}_n| \geq n^{-1/4}) \\ &= \mathbb{P}\big(\sqrt{n}\,|\hat{\theta}_n - \theta_0| \geq n^{1/4}\big). \end{aligned}$$

Recall that by Prop 1 in lecture 4, $\sqrt{n}(\hat{\theta}_n - \theta_0) = O_{\mathbb{P}}(1)$, which implies the RHS $\xrightarrow{n\to\infty} 0$. In other words,

$$\mathbb{P}(\tilde{\theta}_n = 0) \xrightarrow{n\to\infty} 1 \quad \text{or} \quad \sqrt{n}(\tilde{\theta}_n - \theta_0) \xrightarrow{d} 0.$$

Hence the asymptotic variance is $0 < I^{-1}(\theta_0)$ in general.

Mean squared error



By a "minimax" criterion, the Hodges estimator is suboptimal.

End of lecture 7

5. Asymptotic Inference

Thm 4:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)).$$

To construct a confidence set for θ_0, we could use a pivot:

$$I^{1/2}(\theta_0)\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I) \qquad \text{by CMT.}$$

Problem: we don't know θ_0, so can't compute $I(\theta_0)$.

Define the observed Fisher info matrix:

$$J_n(\theta)_{jk} := -\frac{1}{n}\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta_j \partial\theta_k} \log f(X_i, \theta).$$

Lemma (last time) If $\tilde{\theta}_n$ is in the line segment between θ_0 & $\hat{\theta}_n$,

$$J_n(\tilde{\theta}_n) \xrightarrow{P} I(\theta_0).$$

Consider a coordinate $j \in \{1, \ldots, p\}$. By the lemma & CMT, we have

$$\frac{\sqrt{n}(\hat{\theta}_{n,j} - \theta_{0,j})}{\sqrt{\left(J_n(\hat{\theta}_n)^{-1}\right)_{jj}}} \xrightarrow{d} N(0,1).$$

We can use this to define the Wald confidence interval:

$$C_\alpha := \left\{\theta_j \in \mathbb{R} : |\theta_j - \hat{\theta}_{n,j}| \le \frac{z_\alpha}{\sqrt{n}}\sqrt{\left(J_n^{-1}(\hat{\theta}_n)\right)_{jj}}\right\},$$

where z_α is s.t. if $Z \sim N(0,1)$, $\mathbb{P}(|Z| > z_\alpha) = 1 - \alpha$.

We have

$$\mathbb{P}_{\theta_0}(\theta_{0,j} \in C_\alpha) \xrightarrow{n\to\infty} 1 - \alpha.$$

To construct a confidence set for all of θ, we use:

Thm 5 Under regularity, the Wald statistic

$$W(\theta_0) := n(\hat{\theta}_n - \theta_0)^T J_n(\hat{\theta}_n)(\hat{\theta}_n - \theta_0) \xrightarrow{d} \chi^2_p.$$

Proof: By the lemma & CMT, $J_n(\hat{\theta}_n)^{1/2} \xrightarrow{P} I^{1/2}(\theta_0)$. By Slutsky's lemma

$$J_n^{1/2}(\hat{\theta}_n)\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} Z \sim N(0, I).$$

By the CMT for the function $z \mapsto \|z\|^2$, we have that

$$W(\theta_0) = \left\|J_n^{1/2}(\hat{\theta}_n)\sqrt{n}(\hat{\theta}_n - \theta_0)\right\|^2 \xrightarrow{d} \|Z\|^2 \sim \chi^2_p.$$

□

An asymptotic $(1-\alpha)$-confidence set for θ_0 is given by the elliptical region

$$C_\alpha := \{\theta \in \mathbb{R}^p : W(\theta) < \xi_\alpha\},$$

where ξ_α is s.t. if $W \sim \chi^2_p$, $\mathbb{P}(W \ge \xi_\alpha) = \alpha$. Again

$$\mathbb{P}_{\theta_0}(\theta_0 \in C_\alpha) = \mathbb{P}_{\theta_0}(W(\theta_0) < \xi_\alpha) \xrightarrow{n\to\infty} 1 - \alpha.$$

Using the same Wald statistic, we can test the point null hypothesis $H_0 : \theta = \theta_0$. Define the test

$$\psi^{\text{Wald}} = \mathbf{1}_{\{W(\theta_0) > \xi_\alpha\}}.$$

By the same argument, the Type I error probability or size has

$$\mathbb{E}_{H_0}\psi^{\text{Wald}} \xrightarrow{n\to\infty} \alpha.$$

Likelihood ratio & score tests

Let $H_0 : \theta \in \Theta_0$, $H_1 : \theta \in \Theta \subseteq \mathbb{R}^p$, where $\Theta_0 \subseteq \Theta$ is a linear subspace of dimension p_0.

Thm 6 (Wilks' theorem) Under regularity, if $\theta_0 \in \Theta_0$,

$$\hat{\Lambda}_n := 2\left(\underbrace{\sup_{\theta\in\Theta} \ell_n(\theta)}_{\ell_n(\hat{\theta}_n)} - \underbrace{\sup_{\theta\in\Theta_0} \ell_n(\theta)}_{\ell_n(\theta_0)}\right) \xrightarrow{d} \chi^2_{p-p_0}.$$

Proof (Case $\Theta_0 = \{\theta_0\}$, $p_0 = 0$)

Taylor expand $\ell_n(\theta)$ around $\ell_n(\hat{\theta}_n)$

$$2\big(\ell_n(\theta_0) - \ell_n(\hat{\theta}_n)\big) = \underbrace{\frac{\partial \ell_n(\hat{\theta}_n)}{\partial \theta}}_{0}(\theta_0 - \hat{\theta}_n) + \frac{n}{2}(\theta_0 - \hat{\theta}_n)^T J_n(\tilde{\theta}_n)(\theta_0 - \hat{\theta}_n).$$

Where $\tilde{\theta}_n$ is in the line segment between θ_0 & $\hat{\theta}_n$. The same argument in Thm 5 is used to show that the RHS $\xrightarrow{d} \chi^2_p$. □

Wilks' test for $H_0 : \theta \in \Theta_0$ is given by

$$\psi^{\text{Wilks}} = \mathbf{1}_{\{\hat{\Lambda}_n > \xi_\alpha\}},$$

where ξ_α is the upper α point of $\chi^2_{p-p_0}$. By Thm 6, the size

$$\mathbb{E}_{\theta_0}\psi^{\text{Wilks}} \xrightarrow{n\to\infty} \alpha \qquad \forall \theta_0 \in \Theta_0.$$

Note: This test does not require computing $J_n(\hat{\theta}_n)$.

The score test does not even require computing the MLE! (For a point null hypothesis). Define the score test statistic

$$\lambda_n := \frac{1}{n} S_n(\theta_0)^T I^{-1}(\theta_0) S_n(\theta_0).$$

Here $S_n(\theta) = \frac{\partial}{\partial\theta}\ell_n(\theta)$ is the score.

By the Central Limit Theorem

$$\frac{1}{n} S_n(\theta_0) \xrightarrow{d} N(0, I(\theta_0)).$$

By the CMT $\lambda_n \xrightarrow{d} \chi^2_p$ under $H_0 : \theta = \theta_0$.

The score test is

$$\psi^{\text{score}} = \mathbf{1}_{\{\lambda_n > \xi_\alpha\}}.$$

When $H_0 : \theta \in \Theta_0$ is composite ($\dim \Theta_0 > 0$), we can define the score test statistic

$$\lambda_n := \frac{1}{n} S_n(\hat{\theta}^{(0)})^T I^{-1}(\hat{\theta}^{(0)}) S_n(\hat{\theta}^{(0)}),$$

where $\hat{\theta}^{(0)}$ is a "restricted" MLE, that is a maximiser of ℓ_n over $\theta \in \Theta_0$. One can show that under regularity, $\lambda_n \xrightarrow{d} \chi^2_{p-p_0}$ as $n \to \infty$ if $\theta_0 \in \Theta_0$.

End of lecture 8

6. The Delta Method

Let $\hat{\theta}$ be a MLE. We wish to estimate $\phi(\theta_0)$ for some (possibly non-linear, non-injective) function $\phi : \Theta \to \mathbb{R}^m$. A profile likelihood estimator of $\phi(\theta_0)$ maximises

$$L^*(\psi) = \sup_{\substack{\theta \in \Theta \\ \phi(\theta)=\psi}} L(\theta)$$

over $\psi \in \phi(\Theta)$.

Lemma The plug-in MLE, $\phi(\hat{\theta})$, maximises L^*.
Proof Suppose $\exists\, \tilde{\psi}$ s.t.

$$L^*(\tilde{\psi}) > L^*(\phi(\hat{\theta})) + \varepsilon \qquad \text{for some } \varepsilon > 0.$$

By definition of L^*, $\exists\, \tilde{\theta}$ s.t.

$$L(\tilde{\theta}) > L^*(\tilde{\psi}) - \varepsilon > L^*(\phi(\hat{\theta})) \geq L(\hat{\theta}),$$

which contradicts the maximality of the MLE $\hat{\theta}$. □

The Delta Method allows us to characterise the asymptotic distribution of fluctuations $(\phi(\hat{\theta}) - \phi(\theta_0))\sqrt{n}$, which can be used for inference of $\phi(\theta_0)$.

Theorem 7 Let $\hat{\theta}_n$ be an estimator s.t.

$$r_n(\hat{\theta}_n - \theta_0) \xrightarrow{d} Z$$

for some random variable Z and a deterministic rate $r_n \xrightarrow{n\to\infty} \infty$ (e.g. $r_n = \sqrt{n}$). [$\hat{\theta}_n$ need not be the MLE!] Let $\phi : \Theta \to \mathbb{R}^m$ be a continuously differentiable function. Then

$$r_n\big(\phi(\hat{\theta}_n) - \phi(\theta_0)\big) \xrightarrow{d} \begin{pmatrix} \nabla\phi_1(\theta_0)^T \\ \vdots \\ \nabla\phi_m(\theta_0)^T \end{pmatrix} Z \qquad \text{as } n \to \infty.$$

Proof By the mean value theorem,

$$r_n\big(\phi(\hat{\theta}_n) - \phi(\theta_0)\big) = \underbrace{\begin{pmatrix} \nabla\phi_1(\tilde{\theta}_n^{(1)})^T \\ \vdots \\ \nabla\phi_m(\tilde{\theta}_n^{(m)})^T \end{pmatrix}}_{=:D_n} \underbrace{(\hat{\theta}_n - \theta_0)r_n}_{\xrightarrow{d} Z},$$

where $\tilde{\theta}_n^{(i)}$ is on the segment between θ_0 and $\hat{\theta}_n$ for each $i = 1, \ldots, m$. Note that $r_n(\hat{\theta}_n - \theta_0) = O_{\mathbb{P}}(1)$ (by Prop 1(4)); as $r_n \to \infty$, this shows that $\hat{\theta}_n \xrightarrow{P} \theta_0$ (Ex Sheet). Therefore $\tilde{\theta}_n^{(i)} \xrightarrow{P} \theta_0$ for all $i = 1, \ldots, m$. As ϕ is continuously differentiable, by the CMT we have $D_n \xrightarrow{P} \nabla\phi(\theta_0)$. Finally, Slutsky's lemma gives the desired conclusion. □

Notation (supplement): Here the board's $\nabla\phi(\theta_0)$ denotes the Jacobian matrix

$$J_\phi(\theta_0) := \begin{pmatrix} \nabla\phi_1(\theta_0)^T \\ \vdots \\ \nabla\phi_m(\theta_0)^T \end{pmatrix},$$

with one transposed component gradient per row, just as in D_n (with k outputs and m input coordinates, the supplement writes these matrices as elements of $\mathbb{R}^{k\times m}$).

Corollary If $\hat{\theta}_n$ is an asymptotically normal MLE,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)),$$

then

$$\sqrt{n}\big(\phi(\hat{\theta}_n) - \phi(\theta_0)\big) \xrightarrow{d} N\big(0, \nabla\phi(\theta_0)^T I^{-1}(\theta_0)\nabla\phi(\theta_0)\big).$$

Recall that by the Cramér–Rao lower bound for $\phi : \Theta \to \mathbb{R}$, any unbiased estimate $\hat{\phi}$ ($\mathbb{E}_{\theta_0}\hat{\phi} = \phi(\theta_0)$ for all θ_0) has

$$\mathrm{Var}(\hat{\phi}) \geq \frac{1}{n}\nabla\phi(\theta_0)^T I^{-1}(\theta_0)\nabla\phi(\theta_0) \qquad \text{for any finite } n.$$

We see that the plug-in MLE $\phi(\hat{\theta}_n)$ attains or matches this lower bound asymptotically.

Notation (supplement): The board's formulas above are retained: their $\nabla\phi(\theta_0)$ is the transpose of the Jacobian $J_\phi(\theta_0)$ defined in the proof, so the covariance is $J_\phi I^{-1} J_\phi^T$. For scalar ϕ, this is the column gradient, explaining the transpose on the left and its absence on the right in the Cramér–Rao bound.

Example Suppose $(X_i) \overset{\text{iid}}{\sim} N(\mu, 1)$. We wish to estimate μ_0^2 with the plug-in MLE $(\bar{X}_n)^2$. In this case we know

$$\sqrt{n}(\bar{X}_n - \mu_0) \xrightarrow{d} N(0,1) \qquad \text{by the CLT.}$$

By Theorem 7, we have

$$\sqrt{n}(\bar{X}_n^2 - \mu_0^2) \xrightarrow{d} N(0, 4\mu_0^2).$$

When $\mu_0 = 0$, the limit has variance 0 ($\bar{X}_n^2 - \mu_0^2$ converges "faster" than $\sqrt{n}$). We can instead scale the fluctuations $\bar{X}_n^2 - \mu_0^2$ by $r_n = n$:

$$n(\bar{X}_n^2 - 0) = \Big(\underbrace{\sqrt{n}\,\bar{X}_n}_{\xrightarrow{d} N(0,1) \text{ by CLT}} \Big)^2 \xrightarrow{d} \chi_1^2 \qquad \text{by continuous mapping.}$$

II. BAYESIAN INFERENCE

Suppose observation X comes from a statistical model $\{f(\cdot, \theta) : \theta \in \Theta\}$ on a sample space $\mathcal{X}$. We assign a prior distribution π on Θ, which expresses subjective opinions about θ. Then we consider X, θ to be random variables with joint pdf (pmf)

$$\pi(\theta) f(x, \theta) \qquad \text{on } \mathcal{X} \times \Theta,$$

i.e. $\theta \sim \pi$, and $X \mid \theta \sim f(\cdot, \theta)$. The posterior distribution is the conditional distribution of $\theta \mid X$, with pdf

$$\pi(\theta \mid x) = \frac{\pi(\theta) f(x, \theta)}{\displaystyle\int_\Theta \pi(\theta) f(x, \theta)\, d\theta} \qquad \text{"Bayes' rule".}$$

Notation (supplement): The board's $f(x, \theta)$ means $f(x \mid \theta)$, the conditional pdf (or pmf) of X given θ; the joint pdf (or pmf) is $\pi(\theta) f(x, \theta)$.

When $X = (X_1, \ldots, X_n)$ are iid and $f(\cdot, \theta)$ denotes the pdf of a single observation, then

$$\pi(\theta \mid x_1, \ldots, x_n) = \frac{\pi(\theta) \displaystyle\prod_{i=1}^n f(x_i, \theta)}{\displaystyle\int_\Theta \pi(\theta) \prod_{i=1}^n f(x_i, \theta)\, d\theta}.$$

The posterior can be used to make decisions under uncertainty conditional on a fixed observation. We use the principle of maximising expected utility (see von Neumann–Morgenstern theorem). This motivates methods for:

1. Estimation, e.g. let
$$\hat{\theta}_n = \int_\Theta \theta\, \pi(\theta \mid x_1, \ldots, x_n)\, d\theta.$$
2. Uncertainty quantification, e.g. a credible set $\hat{C}_n$:
$$\int_{\hat{C}_n} \pi(\theta \mid x_1, \ldots, x_n)\, d\theta = 1 - \alpha. \qquad (*)$$

For example, if $\Theta = \mathbb{R}$, let
$$\hat{C}_n := [\hat{\theta}_n - R_n,\, \hat{\theta}_n + R_n],$$
with R_n chosen s.t. $(*)$ holds. Note that $\hat{\theta}_n, \hat{C}_n$ are simply functions of $x_1, \ldots, x_n$. But we can regard these as random functions of $X_1, \ldots, X_n \sim P_{\theta_0}$ coming from the model with a true θ_0.

End of lecture 9

Recall that
$$\pi(\theta \mid x_1, \ldots, x_n) \propto \pi(\theta) \prod_{i=1}^n f(x_i, \theta).$$

The posterior can be used for:

1. Estimation: the Bayes estimator (posterior mean)
$$\hat{\theta}_B = \int_\Theta \theta\, \pi(\theta \mid x_1, \ldots, x_n)\, d\theta.$$
2. Uncertainty quantification: a $(1-\alpha)$-credible interval
$$\hat{C}_n = [\hat{\theta}_B - R_n,\, \hat{\theta}_B + R_n], \qquad \int_{\hat{C}_n} \pi(\theta \mid x_1, \ldots, x_n)\, d\theta = 1 - \alpha,$$
where R_n is chosen to satisfy the displayed equality.
3. Model comparison: let $\Theta_0 \subseteq \Theta_1$ and define priors π_0, π_1 on Θ_0, Θ_1, respectively. A Bayes factor is
$$\hat{B}_n = \frac{\displaystyle\int_{\Theta_1} f(x, \theta)\pi_1(\theta)\, d\theta}{\displaystyle\int_{\Theta_0} f(x, \theta)\pi_0(\theta)\, d\theta}, \qquad x = (x_1, \ldots, x_n),$$
where $f(x, \theta)$ denotes the joint likelihood of the sample.

Remark $\hat{\theta}_B, \hat{C}_n, \hat{B}_n$ are functions of a sample $x_1, \ldots, x_n$, which Bayesians regard as fixed. However, we can also regard these quantities as functions of
$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(\cdot, \theta_0)$$
for some "true parameter" θ_0, and study their frequentist properties. For example:

- Is $\hat{\theta}_B$ consistent ($\hat{\theta}_B \xrightarrow{P} \theta_0$), or asymptotically efficient?

- Is $\hat{C}_n$ asymptotically a $(1-\alpha)$-confidence set, i.e.

$$\mathbb{P}_{\theta_0}(\theta_0 \in \hat{C}_n) \xrightarrow{n\to\infty} 1-\alpha?$$

- What are the asymptotic size and power of a test

$$\psi = \mathbf{1}_{\{\hat{B}_n > \xi_{\alpha,n}\}}?$$

Example Let $X_1, \ldots, X_n \mid \theta \overset{\text{iid}}{\sim} N(\theta, 1)$, and place a $N(0,1)$ prior on θ. Then

$$\begin{aligned}\pi(\theta \mid x_1, \ldots, x_n) &\propto \pi(\theta) \prod_{i=1}^{n} f(x_i, \theta) \\ &\propto \exp\left\{-\frac{\theta^2}{2} - \frac{1}{2}\sum_{i=1}^{n}(x_i-\theta)^2\right\} \\ &\propto \exp\left\{-\frac{1}{2/(n+1)}\left(\theta - \frac{\sum_{i=1}^{n} x_i}{n+1}\right)^2\right\}.\end{aligned}$$

We deduce that the posterior distribution is

$$N\left(\frac{n}{n+1}\bar{X}_n,\ \frac{1}{n+1}\right).$$

The Bayes estimator, or posterior mean, is

$$\hat{\theta}_B = \frac{n}{n+1}\bar{X}_n \xrightarrow{P} \theta_0,$$

so it is consistent in a frequentist sense. A $(1-\alpha)$-credible interval is

$$\hat{C}_n = \left[\frac{n}{n+1}\bar{X}_n \ \pm\ \Phi^{-1}\left(1-\frac{\alpha}{2}\right)\frac{1}{\sqrt{n+1}}\right].$$

This is similar to the usual $(1-\alpha)$-confidence interval when n is large. In this case, the prior and posterior are both normal ("conjugacy").

More generally, if the likelihood is in an exponential family,

$$f(x, \theta) = \exp\{\theta^T x - M(\theta)\}\,\mu(x), \tag{$*$}$$

and we choose a conjugate prior

$$\pi(\theta) \propto \exp\{c^T\theta - dM(\theta)\}, \tag{$**$}$$

then the posterior is

$$\pi(\theta \mid x_1, \ldots, x_n) \propto \exp\left\{\theta^T\left(c + \sum_{i=1}^{n} x_i\right) - (d+n)M(\theta)\right\},$$

which is of the same form as the prior. Examples include:

- Bernoulli likelihood, Beta prior;
- Poisson likelihood, Gamma prior.

2. How to choose a prior?

Subjectivist approach The prior should express subjective opinions.

Theorem (Diaconis & Ylvisaker, 1979) Suppose $f(\cdot, \theta)$ is in an exponential family $(*)$, and the prior on θ induces a joint distribution on $\theta, X_1, X_2, \ldots$ with the property that

$$\mathbb{E}[X_2 \mid X_1] = aX_1 + b.$$

Then, under regularity, the prior has to be conjugate, of the form $(**)$ with

$$c = \frac{b}{a}, \qquad d = \frac{1-a}{a}.$$

Remark The prior and likelihood define a joint distribution on $\theta, X_1, X_2, \ldots$. The marginal distribution of $X_1, \ldots, X_n$ is called the prior predictive distribution:

$$\pi(x_1, \ldots, x_n) = \int_\Theta \pi(\theta) \prod_{i=1}^n f(x_i, \theta)\, d\theta.$$

Here $X_1, \ldots, X_n$ are not marginally iid in general. The conditional distribution

$$\pi(x_n \mid x_1, \ldots, x_{n-1}) = \frac{\pi(x_1, \ldots, x_n)}{\pi(x_1, \ldots, x_{n-1})}$$

is called the posterior predictive distribution.

Some subjectivists argue that we can only express opinions about $X_1, \ldots, X_n$, not the parameter.

- How do we justify the exponential-family likelihood? (Pitman–Koopman–Darmois.)
- Why should we assume $X_1, \ldots, X_n \mid \theta$ are iid for some statistical model with parameter θ?

Theorem (de Finetti) Let $(X_i)_{i \geq 1}$ take values in $\{0, 1\}$. Suppose that the prior predictive distribution π is exchangeable, i.e. for all $n \geq 1$, all $(x_1, \ldots, x_n) \in \{0,1\}^n$ and all $\sigma \in S_n$,

$$\pi(x_1, \ldots, x_n) = \pi(x_{\sigma(1)}, \ldots, x_{\sigma(n)}).$$

Then there exists a distribution Π_θ on $[0, 1]$ such that

$$\pi(x_1, \ldots, x_n) = \int_0^1 \underbrace{\prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i}}_{\text{Bernoulli likelihood}}\, \Pi_\theta(d\theta).$$

Theorem (Johnson) Let $(X_i)_{i \geq 1}$ take values in $\{1, 2, \ldots, k\}$, for $k \geq 3$. Suppose π is exchangeable and the posterior predictive distribution satisfies a sufficiency condition:

$$\pi(X_n = i \mid x_1, \ldots, x_{n-1}) = g\left(i, n-1, |\{j < n : x_j = i\}|\right).$$

Then

$$\pi(x_1, \ldots, x_n) = \int_\Delta \prod_{i=1}^n \theta_{x_i}\, \Pi_\theta(d\theta),$$

where

$$\Delta = \left\{ \theta \in \mathbb{R}^k : \theta_i \geq 0,\ \sum_{i=1}^k \theta_i = 1 \right\},$$

and Π_θ is either a Dirichlet$(\alpha_1, \ldots, \alpha_k)$ distribution for some $\alpha_i > 0$, or a point mass at some $p \in \Delta$.

End of lecture 10

Objectivist approach

- Choose a prior that is "uninformative" or "flat", or that maximises the influence of the data on the posterior.
- Construct priors that lead to Bayes estimators and credible sets with good frequentist properties.

Example Let $X_i \mid \theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$. A "flat" prior on $\theta \in (0,1)$ is $\pi_\theta(\theta) = 1$. What if instead we parametrise by

$$\psi = \theta^2 = \mathbb{P}_\theta(X_1 = X_2 = 1)?$$

A flat prior on θ induces a prior on ψ through a change of variables:

$$\pi_\psi(\psi) = \pi_\theta(\theta(\psi)) \left| \frac{d\theta}{d\psi} \right| = \frac{1}{2\sqrt{\psi}}, \qquad 0 < \psi < 1,$$

which is decidedly not flat. Thus "flatness" depends on parametrisation.

Definition (Jeffreys prior) The Jeffreys prior on a parameter θ is given by

$$\pi_{J,\theta}(\theta) \propto \sqrt{\det I_\theta(\theta)},$$

where $I_\theta(\theta)$ is the Fisher information matrix for one observation.

Remark 1 This prior puts more mass on values of θ where the data is more "informative".

Remark 2 Jeffreys prior is invariant under reparametrisation. First consider a model with $\Theta = \mathbb{R}$ and a reparametrisation $\psi = h(\theta)$, where h is a differentiable bijection with differentiable inverse $\theta = \theta(\psi)$. Under the usual regularity conditions, the Fisher information for ψ is

$$\begin{aligned}
I_\psi(\psi) &= -\mathbb{E}_{\theta(\psi)} \left[\frac{\partial^2}{\partial \psi^2} \log f(X, \theta(\psi)) \right] \\
&= -\mathbb{E}_{\theta(\psi)} \left[\frac{\partial}{\partial \psi} \left\{ \frac{d\theta}{d\psi} \frac{\partial}{\partial \theta} \log f(X, \theta(\psi)) \right\} \right] \\
&= -\mathbb{E}_{\theta(\psi)} \left[\frac{d^2\theta}{d\psi^2} \frac{\partial}{\partial \theta} \log f(X, \theta(\psi)) + \left(\frac{d\theta}{d\psi} \right)^2 \frac{\partial^2}{\partial \theta^2} \log f(X, \theta(\psi)) \right] \\
&= \left(\frac{d\theta}{d\psi} \right)^2 I_\theta(\theta(\psi)),
\end{aligned}$$

since the score has mean zero. Hence the Jeffreys prior on ψ is

$$\pi_{J,\psi}(\psi) \propto \left| \frac{d\theta}{d\psi} \right| \sqrt{I_\theta(\theta(\psi))}.$$

This is exactly what we obtain by starting from $\pi_{J,\theta}$ and applying a change of variables.

Remark 3 Often

$$\int_\Theta \sqrt{\det I_\theta(\theta)}\, d\theta = \infty.$$

In this case, we say that $\pi_{J,\theta}$ is an improper prior. However, the posterior

$$\pi(\theta \mid x_1, \ldots, x_n) \propto \pi_{J,\theta}(\theta) \prod_{i=1}^n f(x_i, \theta)$$

can still be a probability density, provided its normalising integral is finite and positive.

Example Let $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ and parametrise by $\theta = (\mu, \sigma)^T$, where $\mu \in \mathbb{R}$ and $\sigma > 0$. The Fisher information is

$$I_\theta(\theta) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 2/\sigma^2 \end{pmatrix}.$$

Thus the Jeffreys prior, as a density with respect to $d\mu\, d\sigma$, is

$$\pi_{J,\theta}(\mu, \sigma) \propto \frac{1}{\sigma^2}.$$

It is constant in μ for each σ, hence improper. The posterior is

$$\pi(\mu, \sigma \mid x_1, \ldots, x_n) \propto \frac{1}{\sigma^n} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \frac{1}{\sigma^2}.$$

One can show that this has a finite integral over $\mu \in \mathbb{R}$, $\sigma > 0$ provided $n \geq 2$ and $x_1 \neq x_2$. More generally, it is enough that $n \geq 2$ and $\sum_{i=1}^n (x_i - \bar{x}_n)^2 > 0$. For fixed σ, the conditional posterior over μ is

$$\mu \mid \sigma, x_1, \ldots, x_n \sim N\left(\bar{x}_n, \frac{\sigma^2}{n} \right).$$

3. Frequentist properties of credible intervals

The Bernstein–von Mises phenomenon states that, under suitable regularity conditions, if

$$X_1, X_2, \ldots \overset{\text{iid}}{\sim} P_{\theta_0}$$

for some true parameter θ_0, then the posterior $\pi(\theta \mid X_1, \ldots, X_n)$, viewed as a random distribution, is approximately

$$N\left(\hat{\theta}_n, \frac{I_\theta(\theta_0)^{-1}}{n} \right), \qquad n \to \infty,$$

where $\hat{\theta}_n$ is the MLE. Concretely, if $\hat{\phi}_n$ is the pdf of this normal distribution, we have

$$\begin{aligned} &2\,\mathrm{TV}\big(\pi(\cdot \mid X_1, \ldots, X_n), \hat{\phi}_n\big) \\ &\quad = \int_\Theta \Big| \pi(\theta \mid X_1, \ldots, X_n) - \hat{\phi}_n(\theta) \Big|\, d\theta \xrightarrow{P_{\theta_0}} 0. \end{aligned} \tag{$*$}$$

Proposition Let $\Theta = \mathbb{R}$, $0 < \alpha < 1$, and define

$$\hat{C}_n = \left\{ \theta : |\theta - \hat{\theta}_n| \leq \frac{R_n}{\sqrt{n}} \right\},$$

where R_n is chosen so that

$$\int_{\hat{C}_n} \pi(\theta \mid X_1, \ldots, X_n)\, d\theta = 1 - \alpha.$$

If the Bernstein–von Mises phenomenon $(*)$ holds and the MLE has its usual asymptotic normal distribution, then

$$\mathbb{P}_{\theta_0}(\theta_0 \in \hat{C}_n) \longrightarrow 1 - \alpha.$$

Thus this credible set is an asymptotic $(1 - \alpha)$-confidence interval.

Proof Let $h = \sqrt{n}(\theta - \hat{\theta}_n)$, let $\widetilde{\pi}_n(h \mid X_1, \ldots, X_n)$ be the posterior density of h, and let $\widetilde{\phi}$ be the density of $N(0, I_\theta(\theta_0)^{-1})$. By definition of $\hat{C}_n$,

$$\int_{-R_n}^{R_n} \widetilde{\pi}_n(h \mid X_1, \ldots, X_n)\, dh = 1 - \alpha.$$

But

$$\begin{aligned}\left| \int_{-R_n}^{R_n} \widetilde{\pi}_n(h \mid X_1, \ldots, X_n)\, dh - \int_{-R_n}^{R_n} \widetilde{\phi}(h)\, dh \right| \\ \leq \int_{-R_n}^{R_n} \left| \widetilde{\pi}_n(h \mid X_1, \ldots, X_n) - \widetilde{\phi}(h) \right| dh \xrightarrow{P_{\theta_0}} 0\end{aligned}$$

by $(*)$ and the change of variables. Write

$$G(t) = \int_{-t}^{t} \widetilde{\phi}(h)\, dh, \qquad t \geq 0.$$

Then $G(R_n) \xrightarrow{P_{\theta_0}} 1 - \alpha$. The function G has a continuous inverse on $(0, 1)$, so the continuous mapping theorem gives

$$R_n \xrightarrow{P_{\theta_0}} q := G^{-1}(1 - \alpha) > 0.$$

By the asymptotic normality of the MLE,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \widetilde{Z}, \qquad \widetilde{Z} \sim N(0, I_\theta(\theta_0)^{-1}).$$

Slutsky's theorem therefore gives

$$\sqrt{n}(\hat{\theta}_n - \theta_0)\frac{q}{R_n} \xrightarrow{d} \widetilde{Z}.$$

Consequently,

$$\begin{aligned}\mathbb{P}_{\theta_0}(\theta_0 \in \hat{C}_n) &= \mathbb{P}_{\theta_0}\left(-q \leq \sqrt{n}(\hat{\theta}_n - \theta_0)\frac{q}{R_n} \leq q \right) \\ &\longrightarrow \mathbb{P}(-q \leq \widetilde{Z} \leq q) \\ &= G(q) = 1 - \alpha.\end{aligned}$$

□

End of lecture 11

Thm (Bernstein–von Mises) Let $\{f(\cdot, \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a statistical model satisfying the reg. cond. for asymptotic normality of the MLE $\hat{\theta}_n$. Let $(X_i) \overset{\text{iid}}{\sim} P_{\theta_0}$ for $\theta_0 \in \text{int}(\Theta)$. Suppose the prior Π is positive & continuous at θ_0, and let

$$\Pi_n(\theta) \propto \Pi(\theta) \cdot \prod_{i=1}^{n} f(X_i, \theta)$$

be the (random) posterior of θ based on a sample $X_1, \ldots, X_n$. Suppose that $\log \Pi_n$ is concave. Let $h = \sqrt{n}(\theta - \hat{\theta}_n)$. Then the posterior distribution of h, $\widetilde{\Pi}_n$, has

$$\int_{\mathbb{R}^d} \left| \widetilde{\Pi}_n(h) - \underbrace{\sqrt{\frac{\det I(\theta_0)}{(2\pi)^d}} \exp\left\{ -\frac{h^T I(\theta_0) h}{2} \right\}}_{\text{pdf of } N(0, I(\theta_0)^{-1})} \right| dh \xrightarrow{P} 0.$$

Proof Let

$$g_n(h) = \log \frac{\widetilde{\Pi}_n(h)}{\widetilde{\Pi}_n(0)} = \log \frac{\Pi_n(\hat{\theta}_n + h/\sqrt{n})}{\Pi_n(\hat{\theta}_n)}$$
$$= \underbrace{\ell_n(\hat{\theta}_n + h/\sqrt{n}) - \ell_n(\hat{\theta}_n)}_{\text{(I)}} + \underbrace{\log \Pi(\hat{\theta}_n + h/\sqrt{n}) - \log \Pi(\hat{\theta}_n)}_{\text{(II)}}.$$

A Taylor expansion of ℓ_n around $\hat{\theta}_n$ (as in Wilks' thm) yields

$$\text{(I)} = -\frac{h^T I(\theta_0) h}{2} + r_n(h),$$

where one can show that for any $R < \infty$,

$$\sup_{\|h\| \le R} |r_n(h)| \xrightarrow{P} 0 \qquad \text{(ULLN)}.$$

For (II), note by consistency of MLE,

$$\left| \hat{\theta}_n + \frac{R}{\sqrt{n}} - \theta_0 \right| \xrightarrow{P} 0.$$

Hence (II) $\xrightarrow{P} 0$, using the fact that $\log \Pi$ is continuous at θ_0.

Hence, writing $q(h) = -\frac{1}{2} h^T I(\theta_0) h$, we've shown

$$\sup_{\|h\| < R} |g_n(h) - q(h)| \xrightarrow{P} 0.$$

Note, $g_n(h)$ is concave, and $g_n(0) = 0$.

Claim.

$$\int_{\mathbb{R}^d} \left| e^{g_n(h)} - e^{q(h)} \right| \xrightarrow{P} 0.$$

The BvM theorem we want to prove says that

$$\int \left| \frac{e^{g_n(h)}}{Z_n} - \frac{e^{q(h)}}{Z} \right| dh \xrightarrow{P} 0,$$

where

$$Z_n = \int e^{g_n(h)}\, dh, \qquad Z = \int e^{q(h)}\, dh = \sqrt{\frac{(2\pi)^d}{\det I(\theta_0)}}.$$

By the triangle ineq.

$$|Z_n - Z| \le \int \left| e^{g_n(h)} - e^{q(h)} \right| dh \xrightarrow{P} 0.$$

Then,

$$(*) \le \frac{1}{Z_n} \underbrace{\int \left| e^{g_n(h)} - e^{q(h)} \right| dh}_{= A_n} + \left| \frac{1}{Z_n} - \frac{1}{Z} \right| \int e^{q(h)}\, dh \qquad (\Delta \text{ ineq.})$$
$$\le \frac{A_n}{Z_n} + \frac{|Z_n - Z|}{Z_n} \le \frac{2A_n}{Z_n} \xrightarrow{P} 0 \quad \text{by continuous mapping.}$$

This proves the result. It remains to show the claim holds.

1. Build an envelope for $\exp(g_n)$

Since $I(\theta_0)$ is pos. def., $\exists \lambda > 0$ s.t.

$$h^T I(\theta_0) h \geq \lambda \|h\|^2.$$

Therefore on the unit sphere $\|h\| = 1$,

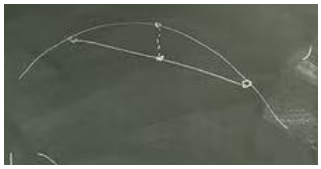
$$q(h) \leq -\frac{\lambda}{2}.$$

On an event E_n with $\mathbb{P}(E_n) \to 1$,

$$g_n(u) \leq -\frac{\lambda}{4} \qquad \text{for all } \|u\| = 1.$$

Fix unit vector $\|u\| = 1$, let $r > 1$, by concavity

$$\begin{aligned} g_n(u) &= g_n\left(\frac{1}{r}(ru) + \left(1 - \frac{1}{r}\right) 0\right) \\ &\geq \frac{1}{r} g_n(ru) + \left(1 - \frac{1}{r}\right) \underbrace{g_n(0)}_{0} \qquad \text{(concavity)}. \end{aligned}$$



Hence, on E_n,

$$g_n(ru) \leq r g_n(u) \leq -\frac{r\lambda}{4}$$

and

$$e^{g_n(h)} \leq \exp\left(-\frac{\lambda}{4}\|h\|\right) \qquad \text{for } \|h\| \geq 1.$$

Now for $\|h\| \leq 1$ we have

$$\sup_{\|h\| \leq 1} |g_n(h) - q(h)| \xrightarrow{P} 0, \qquad \text{where } q(h) = -\frac{h^T I(\theta_0) h}{2} \leq 0.$$

Hence on some E_n' with $\mathbb{P}(E_n') \to 1$,

$$e^{g_n(h)} \leq e^1 \qquad \text{for all } \|h\| \leq 1.$$

On $E_n'' = E_n \cap E_n'$ with $\mathbb{P}(E_n'') \to 1$, we have

$$e^{g_n(h)} \leq H(h) := \begin{cases} e & \text{for } \|h\| \leq 1, \\ e^{-\frac{\lambda}{4}\|h\|} & \text{for } \|h\| > 1. \end{cases}$$

Also $e^{q(h)} \leq H(h)$.

2. Split the integral

$$\int \left| e^{g_n(h)} - e^{q(h)} \right| dh \leq \underbrace{\int_{\|h\| \leq R} |\cdot|}_{\text{(I)}} + \underbrace{\int_{\|h\| > R} |\cdot| \, dh}_{\text{(II)}}.$$

Since $H(h)$ is integrable, we can pick R s.t.

$$2\int_{\|h\|>R} H(h) < \frac{\varepsilon}{2}.$$

Hence on E_n'', (II) $< \frac{\varepsilon}{2}$. For (I) use

$$\sup_{\|h\|\le R} |g_n(h) - q(h)| \xrightarrow{P} 0.$$

In particular,

$$\begin{aligned}(\mathrm{I}) &= \int_{\|h\|\le R} \left| e^{g_n(h)} - e^{q(h)} \right| dh \\ &\le \mathrm{Vol}(\{\|h\| \le R\}) \cdot \sup_{\|h\|\le R} \left| e^{g_n(h)} - e^{q(h)} \right| \xrightarrow{P} 0\end{aligned}$$

as the function $x \mapsto \exp(x)$ is uniformly continuous on compact sets. So on some event E_n''', with $\mathbb{P}(E_n''') \to 1$, (I) $< \frac{\varepsilon}{2}$.

$$\therefore \quad \mathbb{P}\left(\int \left| e^{g_n(h)} - e^{q(h)} \right| < \varepsilon \right) \ge \mathbb{P}(E_n''' \cap E_n'') \longrightarrow 1.$$

□

Rmk. We used concavity to control tails. Without concavity need to show posterior concentrates on neighbourhood of $\hat{\theta}_n$ where quadratic approx. holds.

Many log-posteriors are concave. In particular, if $f(\cdot, \theta)$ is a natural exponential family,

$$f(x, \theta) = \exp\{\theta^T x - M(\theta)\}\, g(x),$$

one can show that $\nabla^2 M(\theta) \succeq 0$, hence $\log f(x, \theta)$ is concave. The conjugate prior

$$\Pi(\theta) \propto \exp\{b^T\theta + dM(\theta)\}, \qquad d \le 0,$$

is also log-concave.

End of lecture 12